

```

1: `timescale 1ns / 1ps
2: //////////////////////////////////////////////////
3: // Company:
4: // Engineer:
5: //
6: // Create Date: 05/25/2026 01:55:07 PM
7: // Design Name:
8: // Module Name: ahb_pmu
9: // Project Name:
10: // Target Devices:
11: // Tool Versions:
12: // Description:
13: //
14: // Dependencies:
15: //
16: // Revision:
17: // Revision 0.01 - File Created
18: // Additional Comments:
19: //
20: //////////////////////////////////////////////////
21:
22:
23: module ahb_pmu(
24:     // AHB Interface
25:     input wire hclk,
26:     input wire hresetn,
27:     input wire hsel,
28:     input wire [31:0] haddr,
29:     input wire [31:0] hwdata,
30:     input wire hwrite,
31:     input wire [1:0] htrans,
32:
33:     output wire [31:0] hrdata,
34:     output wire hreadyout,
35:
36:
37:     //PMU Specific Outputs
38:     output wire uart_clk_en,
39:     output wire spi_clk_en,
40:     output wire i2c_clk_en
41: );
42:
43: // The single Power/Clock Control Register
44: reg [31:0] clk_enable_reg;
45:
46: always @(posedge hclk or negedge hresetn) begin
47:     if(!hresetn) begin
48:         // Default to 0 (Everything is OFF on boot)
49:         clk_enable_reg <= 12'h00000000;
50:     end else begin
51:         //Single-cycle write catch (same as our SPI fix!)
52:         if(hsel && htrans == 2'b10 && hwrite) begin
53:             if(haddr[7:0] == 8'h00) begin
54:                 clk_enable_reg <= hwdata;
55:             end
56:         end
57:     end
58: end
59:
60: // Map the register bits to the output wires
61: assign uart_clk_en = clk_enable_reg[0];
62: assign spi_clk_en = clk_enable_reg[1];
63: assign i2c_clk_en = clk_enable_reg[2];
64:
65: //CPU read logic
66: assign hrdata = (hsel && !hwrite && haddr[7:0] == 8'h00) ? clk_enable_reg : 32'h0;
67:
68: //We never stall the bus
69: assign hreadyout = 1'b1;
70:
71: endmodule
72:

```